

3139544 - Optimize Service Layer Performance

Component: SBO-SDK-SVL-PER (SAP Business One > Software Development Kit > Service Layer > Service Layer Performance), Version: 6, Released On: 24.02.2025

Symptom

You are implementing applications based on Service Layer, you are looking for guidelines for optimal implementation.
(or)

You are using applications based on Service Layer and you are facing slowness.

Cause

System behavior/ Functionality description/ Business process

Solution

The standard approach is to find the below parameters for the implementation:

- **Number of concurrent requests** - Requests that are coming into service layer at the same time. Consider all components which can generate request.
 - Job service - Alert and Scheduling
 - Electronic Document Service (*EDS*)
 - Web Client
 - SAP Business One Integration (B1i) scenarios using Service Layer
 - Mobile service
 - All partner add-ons using service layer
- **Total requests** - Total number of requests processed by Service Layer
- **Type of requests** - (get/patch/put/post)
- **Complexity of the request**
 - What are the objects used in request(Like Documents, Items, Production Order, Picklists, Activity, Contacts..)
 - How many lines are processed in your requests
 - How many serial number and batch numbers are processed.
 - How many bin locations are processed.
- **Database response time**
 - Size of the database.
 - Number of rows on the table you operate.
 - Number of other concurrent processes which could generate database activity.(Other add-ons, Business One Client, reports, alerts and so on).

Based on the above parameters, you can find the suitable configuration and arrive at a baseline for your implementation.

- **Deciding load balancer members/node**
 - Service Layer load balancer uses the “*bybusyness*” scheduling algorithm. It distributes requests to members having least active connections.
 - Members can be added/removed using [ServiceLayerController](#) or using installer.
 - - Find out how many schemas (for example, 2), and how many concurrent distinct users (for example, 8) per schema the customers have. Then install Service Layer with $2 \times 8 = 16$ load balancer member; the minimum memory needed is 16GB.
 - Find out high priority requests and Isolate repeated requests into a separate node, refer to KBA:3311536.
- **Deciding on number of threads(Maximum Threads per Load Balancer Member)**
 - This setting “Maximum Threads per Load Balancer Member” will generate as many child process as set for each load balancer member. Refer online [help](#).
 - For Service Layer on HANA, you can specify the Maximum Threads per Load Balancer Member during the installation. This setting will generate as many as child process for each load balancer member. For example: Maximum Threads per Load Balancer Member is set to 20 and you have 4 load balancer members. Then, a maximum of $20 \times 4 = 80$ child process can be reached. It can consume up to 80 GB of memory.
 - For Service Layer on Windows, you cannot change Maximum Threads per Load Balancer Member during the installation. By default, it is set as 1. Internally threads are handled by the windows operating system.
 - The default setting in the installer is 24, this should be sufficient for most of the scenarios. If the same user makes multiple requests from different applications, then such requests will be processed by the same nodes; hence, in such a case you would need more than

one thread per node. Refer to Apache [documentation](#).

- Each active thread handling complex requests can consume up to 1 GB of RAM. Manage the threads based on available RAM.

• Deciding on maximum connection per child

- This setting “Max Connections Per Child” limits the number of connections that an individual child server process will handle, once the limit is reached the child process will close.
- This setting can be updated using [ServiceLayerController](#).
 - Set a value of 1024 or higher. This ensures that the child process will run for an extended period, keeping all loaded data available in the buffer, which enhances performance. This setting is particularly useful for machines with low memory specifications.

• Deciding on sticky session

- “Sticky sessions”, also known as session persistence, once a user is directed to a specific member, all subsequent requests from that user are routed to the same member. It ensures a seamless and personalized experience for every single user.
- This setting can be updated using [ServiceLayerController](#).
 - Enable sticky sessions for non-concurrent load.
 - Disable sticky sessions for concurrent load.

• Deciding on the deployment type

- The Service Layer can operate on a separate machine and does not have to be hosted on the same machine as the database server or the SLD (System Landscape Directory).
- Split the load by installing separate Service Layer instance based on configuration e.g.: One instance using sticky session and another instance without sticky session.
 - You can install one instance of service for all the SAP Business One Components like (EDS, JobService, Wbeclient) and another separate instance for all third party applications.

• Maintenance - Recommendations

- Restart the Service Layer and System Landscape Directory (SLD) services at least once a day.
 - For Service layer on Linux, refer [\[crontab\]](#) to schedule restart.
 - For service layer on Windows, Use Windows Task Scheduler to schedule the script to run periodically. Refer to the [Microsoft Task Scheduler](#) documentation.
- Clear unnecessary logs and core dumps regularly.
 - Clear unnecessary logs or core dumps regularly. Refer SAP Note: [2441140](#) for log location.
 - Debug or detailed logs consumes considerable disk space. Do not enable any detailed logs or debug logs unless you are investigating an issue. Refer KBA: [3543181,3538414](#).
- Perform database maintenance tasks regularly.
 - For SAP Business One, version for SAP HANA, refer to SAP Note: [2540510](#).
 - For Microsoft SQL Server, refer to SAP KBA: [3107985](#) and refer to SAP Note [1241422](#).
- Plan bulk upload operations in lean hours.
 - When SAP HANA database schema and Service Layer are running on the same machine, make sure that standard backup guidelines are followed. Refer to SAP Note [1950261](#).

• SAP Business One Configurations - Recommendations

- Optimize the line items.
- Keep the number of line items optimal in the documents.
- Optimal usage of serial and batch line items in the document.
- Remove unused bin location and UOM (Unit Of Measures).
- Optimize the history table settings. Refer the online help [\[History setting\]](#) and [\[Log cleanup\]](#).
- Optimize the alerts usage and clear the unused alerts.
- Remove the unused pricelists, warehouse. Refer to SAP Note: [3135722](#).
- Use archiving functionality whenever possible.
- Define default series for documents created in bulk processing.

• Development – Recommendations

- Avoid repeated login logic and reuse the same sessions. Resource consumption (RAM, CPU) increases with each login request. The login session can be reused until it expires, the session timeout can be managed in the [ServiceLayerController](#). Refer to online [help](#).
- Use properties only which are required to maintain optimal payload size.
- Use ASYNC for GET calls and SYNC for POST/PATCH/DELETE calls.
- Use appropriate filters for the notification procedures and avoid long time consuming logic.
- Use [pagination](#) to load only required content in chunks for the user.
- Use [return no content](#) to minimize data returned on bulk operations.
- Use [etag](#) for optimistic concurrency control for update transactions.
- Use [navigation and association](#) to retrieve data from related sub objects.
- Use [user defined schema](#) as a filter to reduce the data output.
- Use \$batch for short multiple transaction. Avoid long \$batch operations to avoid database blocks.
- Send all request to load balancer port. Directly sending requests to member port will compromise security and will reduce efficiency.

See Also:

- Enablement session: [\[English\]](#) [\[Spanish\]](#).
- FAQ: [Webinar Q&A](#).
- For general performance issues, refer SAP Note: [2617060](#).

- To track the isolated request time, you can enable performance logs to troubleshoot the issue. Refer to [KBA:315798](#).

Products

Products

SAP B1 VERSION FOR SAP HANA all versions

SAP BUSINESS ONE all versions

This document refers to

SAP Note/KBA	Component	Title
3157498		SBO-SDK-SVL
2617060	SBO-SDK-DEV	Troubleshooting Performance Issues When Working with SAP Business One SDK
2441140	SBO-CRO-SUP	Central Note for Log File Locations of SAP Business One Components

This document is referenced by

SAP Note/KBA	Component	Title
3515638		Overview Note for SAP Business One 10.0 SP 2411, version for SAP HANA
3515663	SBO-BC	Overview Note for SAP Business One 10.0 SP 2411